Ingest constructors.jason files

Step 1 - Read the Jason file using Spark dataframe reader

```
dbutils.widgets.text("p_data_source", "")
v_data_source = dbutils.widgets.get("p_data_source")
```

```
dbutils.widgets.text("p_file_date", "2021-03-21")
v_file_date = dbutils.widgets.get("p_file_date")
```

```
%run "../includes/configuration"
```

```
%run "../includes/common_functions"
```

```
constructors_schema = "constructorId INT, constructorRef STRING, name STRING, nationality STRING, url STRING"
```

```
constructor_df = spark.read \
.schema(constructors_schema) \
.json(f"{raw_folder_path}/{v_file_date}/constructors.json")
```

```
# display(constructor_df)
```

Step 2 - Drop unwanted columns from the dataframe

```
from pyspark.sql.functions import col
```

```
constructor_dropped_df = constructor_df.drop(col('url'))
```

```
# display(constructor_dropped_df)
```

Step 3 - Rename columns and add ingestion date

```
from pyspark.sql.functions import current_timestamp, lit
```

```
constructor_renamed_df = constructor_dropped_df.withColumnRenamed("constructorId", "constructor_id") \
                                               .withColumnRenamed("constructorRef", "constructor_ref") \
                                               .withColumn("data_source", lit(v_data_source)) \
                                               .withColumn("file_date", lit(v_file_date))
```

```
constructor_final_df = add_ingestion_date(constructor_renamed_df)
```

```
# display(constructor_final_df)
```

Step 4 Write output to parquet file

```
# constructor_final_df.write.mode("overwrite").format("parquet").saveAsTable("f1_processed.constructors")
```

```
constructor_final_df.write.mode("overwrite").format("delta").saveAsTable("f1_processed.constructors")
```

```
%fs
ls /mnt/formula1dl1204/processed/constructors
```

Table

	path	name
1	dbfs:/mnt/formula1dl1204/processed/constructors/_delta_log/	_delta_log/
2	dbfs:/mnt/formula1dl1204/processed/constructors/part-00000-bcdf1ec2-45a6-4646-a177-e1ee9a9f18c6-c000.snappy.parqu...	part-00000-bcdf1ec2-45a6-4646-a17

2 rows

```
# df = spark.read.parquet(f"{processed_folder_path}/constructors")
```

```
# display(df)
```

```
%sql
desc extended f1_processed.constructors
```

Table

	col_name	data_type
1	constructor_id	int
2	constructor_ref	string
3	name	string
4	nationality	string
5	data_source	string
6	file_date	string
7	ingestion_date	timestamp
8		
9	# Delta Statistics Columns	
10	Column Names	name, ingestion_date, constructor_ref, constructor_id, file_date, nationality, data_source
11	Column Selection Method	first-32
12		
13	# Detailed Table Information	
14	Catalog	spark_catalog

25 rows

```
%sql
select * from f1_processed.constructors;
```

Table

	constructor_id	constructor_ref	name	nationality	data_source	file_date	ingestion_d
1	1	mclaren	McLaren	British		2021-03-21	2024-07-24T14:
2	2	bmw_sauber	BMW Sauber	German		2021-03-21	2024-07-24T14:
3	3	williams	Williams	British		2021-03-21	2024-07-24T14:
4	4	renault	Renault	French		2021-03-21	2024-07-24T14:
5	5	toro_rosso	Toro Rosso	Italian		2021-03-21	2024-07-24T14:
6	6	ferrari	Ferrari	Italian		2021-03-21	2024-07-24T14:
7	7	toyota	Toyota	Japanese		2021-03-21	2024-07-24T14:
8	8	super_aguri	Super Aguri	Japanese		2021-03-21	2024-07-24T14:
9	9	red_bull	Red Bull	Austrian		2021-03-21	2024-07-24T14:
10	10	force_india	Force India	Indian		2021-03-21	2024-07-24T14:
11	11	honda	Honda	Japanese		2021-03-21	2024-07-24T14:
12	12	spyker	Spyker	Dutch		2021-03-21	2024-07-24T14:
13	13	mf1	MF1	Russian		2021-03-21	2024-07-24T14:
14	14	spyker_mf1	Spyker MF1	Dutch		2021-03-21	2024-07-24T14:

211 rows

```
dbutils.notebook.exit("Success")
```

```
Success
```